Automatic Clothing Category Recognition: Handcrafted Features versus Raw Pixels with Classical Machine Learning (Phase 1 — Machine-Learning Evaluation)

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***Abstract*—We study automatic tagging of clothing images for a small catalogue using the Fashion-MNIST benchmark (70,000 grayscale 28×28 images, 10 classes). Phase 1 asks whether a fixed image descriptor, the Histogram of Oriented Gradients (HOG), improves classical classifiers over flattened pixels. We compare Logistic Regression, k-Nearest Neighbours, Random Forest and an RBF Support Vector Machine on three representations (raw pixels, HOG, HOG plus 16 shape statistics) under a strict protocol: the official train/test split is preserved, a stratified validation set is drawn only from the training data, hyper-parameters are tuned on validation, and the test set is used once. HOG improves three of four classifiers significantly (McNemar $p < 0.001$): +4.0 points for Logistic Regression, +2.1 for k-NN and +1.4 for SVM, whereas Random Forest is unchanged (+0.2, $p = 0.49$). The best model, an RBF SVM on HOG, reaches 91.3% test accuracy and 0.913 macro-F1. About 60–65% of all errors of every model stay inside the group {T-shirt, Pullover, Coat, Shirt}, and Shirt recall is only 0.75, which motivates the CNN studied in Phase 2.**

***Index Terms*—Fashion-MNIST, HOG, feature engineering, support vector machine, random forest, k-nearest neighbours, logistic regression, error analysis**

I. Introduction

Online retailers and small shops maintain catalogues in which every product image must carry a category tag. Manual tagging is slow and inconsistent, so an automatic classifier is attractive. Fashion-MNIST [1] was proposed as a drop-in, harder replacement for the digit benchmark MNIST [2]: the images have the same format but represent clothing, and simple classifiers no longer saturate.

Problem statement. Given a 28×28 grayscale image, predict one of ten categories. The project has two phases. Phase 1 (this report) covers the classical machine-learning part: a baseline of flattened pixels and HOG descriptors with four classical learners. Phase 2 will add a small convolutional neural network (CNN) and compare it with Phase 1.

Gap addressed. Flattening an image into a vector discards the 2-D neighbourhood structure, while HOG fixes the representation (local gradient orientations) before the model has seen any category-specific example. It is therefore not obvious which choice is better for which learner, nor at what cost in model size and inference time.

Contributions.

1) A structured data-quality audit and EDA that predicts which classes will be confused (Section III).
2) A feature-engineering study with a HOG hyper-parameter ablation, 16 interpretable shape statistics and PCA (Section IV).
3) A fair, fully reproducible comparison of 4×3 model/feature configurations with accuracy, macro-F1, per-class recall, parameter count and latency (Section VII).
4) Statistical validation (McNemar test, bootstrap confidence intervals) and an error analysis of the upper-body garment group (Section VIII).

II. Literature Review

Benchmark. Xiao et al. [1] released Fashion-MNIST with 60,000 training and 10,000 test images from Zalando articles, converted to 28×28 grayscale, and benchmarked 129 classifier configurations from scikit-learn. They report that tree ensembles and SVMs on raw pixels reach the high 80s to about 90%, and that the shirt-like classes are the hardest. LeCun et al. [2] established the MNIST protocol and showed that convolutional networks trained with gradient-based learning outperform fixed feature extractors on digits.

Handcrafted descriptors. Dalal and Triggs [3] introduced HOG for pedestrian detection: image gradients are binned by orientation inside small cells and normalised over overlapping blocks, giving robustness to local illumination and small shifts. Related descriptors include SIFT [4] and Local Binary Patterns [5]. For small 28×28 inputs the choice of cell size is critical, because a classic 8×8 cell would cover almost a third of the image.

Classical learners. Logistic Regression is the standard linear probabilistic classifier [6]. k-NN [7] is non-parametric and stores the training set. Random Forests [8] average decorrelated decision trees. Support Vector Machines [9] maximise the

TABLE I
DATA-QUALITY AUDIT.

Check	Result
Train / test shape	$(60000, 28, 28)$ / $(10000, 28, 28)$
dtype, range	uint8, $[0, 255]$
Missing values	0
Blank images	0
Exact duplicates (train)	0
Train–test overlap	0
Duplicate images with different labels	0
Zero-valued pixels	50.2%
Mean / std of pixel (scaled)	0.286 / 0.353

margin and, with an RBF kernel, learn non-linear boundaries with implicit infinite-dimensional features.

Dimensionality reduction and visualisation. PCA [10] and t-SNE [11] are used in the EDA to measure redundancy and inspect class overlap.

Evaluation methodology. Because all classifiers are scored on the same 10,000 test images, paired tests are appropriate; McNemar's test [12] on discordant pairs is recommended by Dietterich [13] for single-split comparisons. The bootstrap [14] provides confidence intervals.

Positioning. Prior work either reports raw-pixel baselines or CNNs. Our study isolates the effect of a *fixed* descriptor on four learner families, with equal tuning budget, a clean validation protocol and costs (parameters, latency) reported next to accuracy.

III. Dataset Quality and Exploratory Data Analysis

A. Data description and audit

The raw IDX files are read with `gzip` and `numpy.frombuffer` (`src/data_loader.py`). Table I lists the audit performed by `eda.quality_report`.

The data are clean, so no imputation, removal or de-duplication is required. The only preprocessing step is the scaling $x \leftarrow x/255$. Labels are never altered.

B. Class balance and split shift

Every class has exactly 6,000 training and 1,000 test images (Fig. 1). Accuracy is therefore meaningful, but we still report macro-F1 and per-class recall. A two-sample Kolmogorov–Smirnov test on the per-image mean intensity finds no train/test shift ($D = 0.0064$, $p = 0.87$); the largest mean-pixel difference is 0.0086.

C. Visual and intensity analysis

Fig. 2 shows random samples and Fig. 3 the per-class mean and standard-deviation images. Footwear classes have small bounding-box height and large width (Sandal mean width 27.8 px, height 16.1 px; Sneaker height 13.4 px), Trouser is narrow (11.7 px), and the four upper-body classes have almost identical extents (≈21 px wide, ≈28 px high). The pixel-level standard deviation is highest at garment edges and sleeves. Fig. 4 shows the distribution of brightness, ink coverage and width.

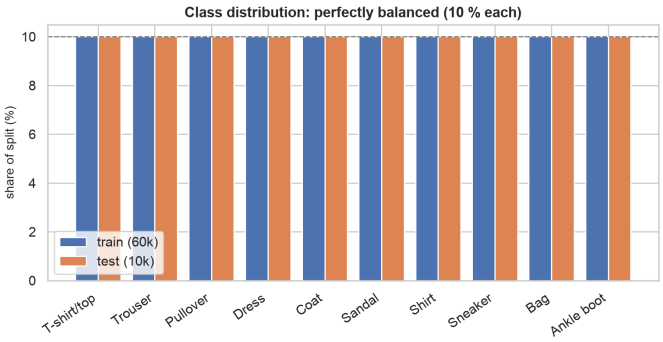


Fig. 1. Balanced classes in both splits.



Fig. 2. Random samples per class.

D. Pixel relevance and spatial structure

`sklearn.feature_selection.f_classif` gives an ANOVA F-score per pixel; together with the pixel variance (Fig. 5) it shows that 31 border pixels have variance below 10^{-3} and carry no information. We then compute the mean Pearson correlation between pixel pairs as a function of their distance (Fig. 6): adjacent pixels correlate at $r \approx 0.75$, decaying to 0.09 at 14 px, whereas the same pixels with shuffled positions stay at $r \approx 0.16$. Neighbourhood structure is therefore real and is ignored by flattening, which is the empirical motivation for HOG.

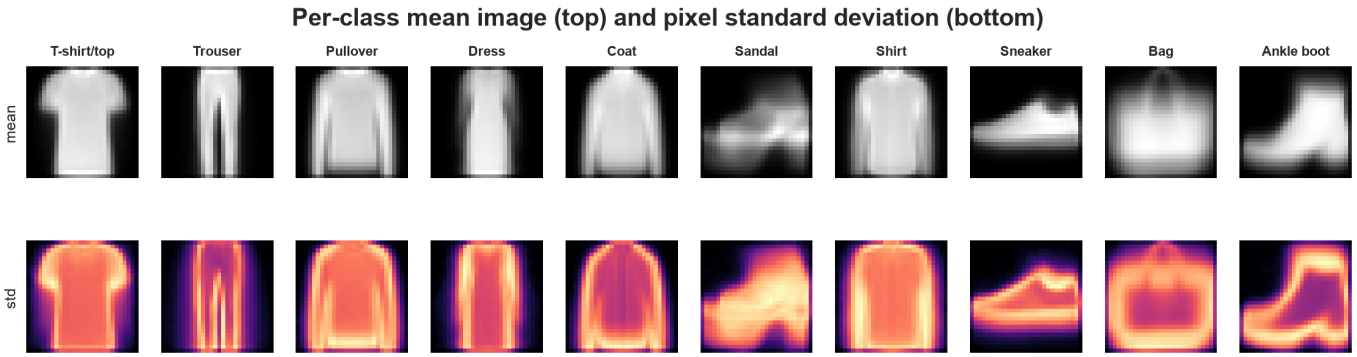


Fig. 3. Class-mean (top) and pixel standard-deviation (bottom) images.

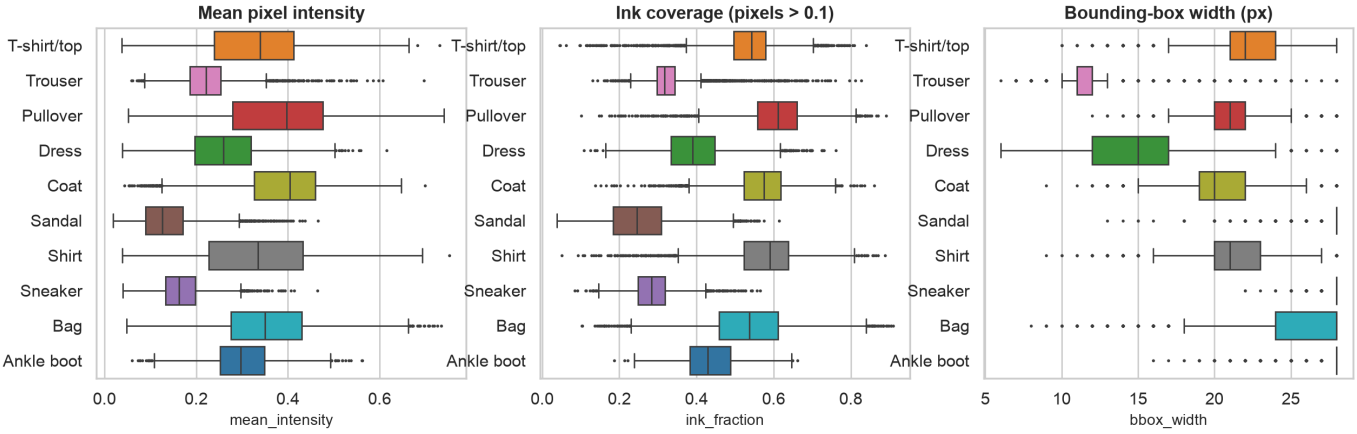


Fig. 4. Per-class distributions of mean intensity, ink coverage and bounding-box width.

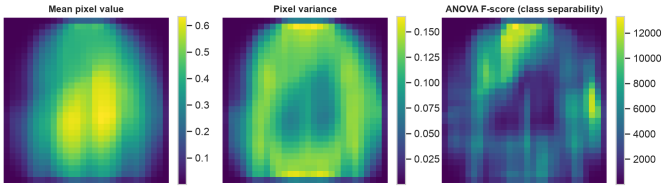


Fig. 5. Mean, variance and F-score per pixel.

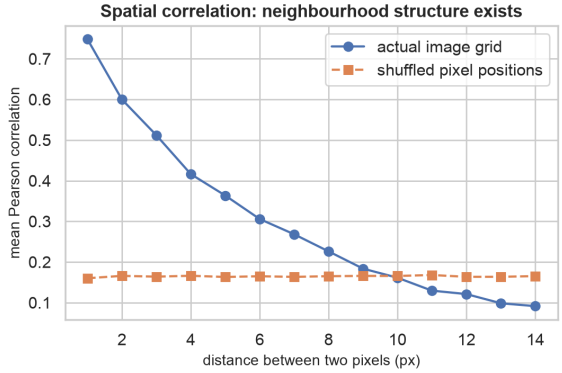


Fig. 6. Pixel correlation versus distance.

E. Class similarity, PCA and t-SNE

The Euclidean distance between class-mean images (Fig. 7) ranks the closest pairs: Pullover–Coat (2.48), Pullover–Shirt (2.68), Coat–Shirt (3.02), T-shirt–Shirt (3.72). Sandal–Sneaker (3.84) is the only non-upper-body pair in this group. This *predicts* the confusion structure later observed in Section VIII.

PCA on pixels (Fig. 8) needs 24, 84 and 187 components to retain 80%, 90% and 95% of the variance, i.e. the 784 pixels are highly redundant. t-SNE on 5,000 images after PCA-50 (Fig. 9) shows compact clusters for Trouser, Bag and footwear and a large overlap among the upper-body classes. Fig. 10 lists examples of that ambiguous group.

F. Orientation check

HOG is orientation-sensitive, so we verified image orientation. The mean top-minus-bottom ink share is positive for T-shirt (+0.08) and Trouser (+0.15) and strongly negative for Sandal, Sneaker, Bag and Ankle boot (≈ -0.3), consistent with upright garments and footwear soles at the bottom. Images are stored row-major, upright, so HOG is applied directly with no rotation.

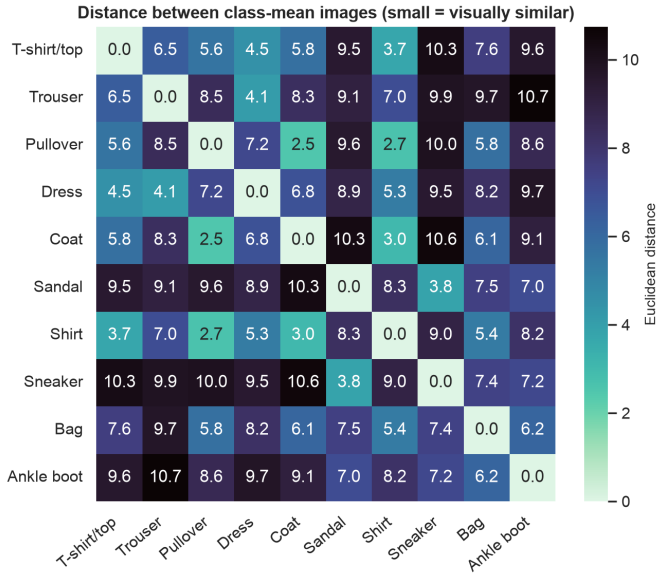


Fig. 7. Distance between class-mean images.

IV. Feature Engineering

Three feature sets are built by `src/features.py`.

A. Raw pixels (784-d)

`flatten_pixels` reshapes the 28×28 image in row-major order after scaling to $[0, 1]$.

B. Histogram of Oriented Gradients

HOG [3] is computed with `skimage.feature.hog`. For each pixel the gradients g_x, g_y give magnitude $m = \sqrt{g_x^2 + g_y^2}$ and orientation $\theta = \operatorname{atan2}(g_y, g_x)$ (unsigned, $[0°, 180°)$). The image is divided into cells of $p \times p$ pixels; each cell accumulates a histogram of B orientation bins weighted by m. Cells are grouped into overlapping blocks of 2×2 cells and each block vector v is normalised with L2-Hys ($v \leftarrow v/\sqrt{\|v\|_2^2 + \epsilon^2}$, clip at 0.2, renormalise). The descriptor is the concatenation of all block vectors. For an $n \times n$ image with $c = n/p$ cells per side the length is $(c-1)^2 \cdot 4 \cdot B$. Fig. 11 visualises the descriptor. Extraction is parallelised with `joblib.Parallel`.

Ablation. We evaluated $B \in \{6, 9, 12\}$, cell size $\in \{2, 4, 7\}$ and block size $\in \{1, 2\}$ cells (36 configurations) with a fast Logistic-Regression probe (`StandardScaler`, $C = 0.5$) trained on 12,000 images and scored on the validation set (Fig. 12). Small cells win (best: $B = 12$, 2×2 cells, 2×2 blocks, 0.870 but 8,112-d), and 2×2 blocks beat 1×1. Since kNN, SVM and RF scale with dimensionality, we select the best configuration with at most 2,000 dimensions: $B = 6$, 4×4 cells, 2×2 blocks, **864-d** (probe accuracy 0.856). This *a priori* fixed descriptor is then used by all four classifiers, matching the project premise that HOG fixes the representation before the model sees category-specific examples.

C. Shape statistics (16-d)

`shape_stats` computes interpretable global descriptors: mean and standard deviation of intensity, ink fraction (pixels > 0.1), bounding-box height, width and aspect ratio, centroid row/column, left–right and top–bottom symmetry ($1 - \|I - \text{flip}(I)\|_1/(2\|I\|_1)$), Sobel edge density (`scipy.ndimage.sobel`), top/bottom-half ink mass, width of the top and bottom rows, and the gap between legs. Each is divided by a fixed constant to bring it to roughly $[0, 1]$ so that it does not dominate the HOG block in distance/kernel models. Fig. 13 shows the class profile; a Random Forest ranks ink fraction, aspect ratio and bottom width highest. Concatenated with HOG they form the `hog_stats` set (880-d).

D. PCA reference and probe comparison

PCA (`sklearn.decomposition.PCA`) with 100–200 components gives 0.857–0.862 probe accuracy, equal to using all 784 pixels (0.855), confirming redundancy but also showing that a linear projection cannot add new information (Fig. 14). Under the same probe, pixels reach 0.855, HOG 0.884 and HOG+stats 0.889 (validation).

V. Methodology: Models and Theory

All models are from `scikit-learn` [15]; `scikit-image` [16] provides HOG.

Logistic Regression (multinomial, `LogisticRegression`, L-BFGS, max 300 iterations). It minimises $-\sum_i \log \text{softmax}(Wx_i + b)_{y_i} + \frac{1}{2C}\|W\|_2^2$. Inputs are standardised (`StandardScaler` inside a `Pipeline`, fitted on the training data only). Parameters: $10 \cdot d + 10$.

k-Nearest Neighbours (`KNeighborsClassifier`). Predicts the (distance-weighted) majority label of the k closest training images in Euclidean distance. Non-parametric: the stored matrix is the model.

Random Forest (`RandomForestClassifier`, 300 trees, $\sqrt{d}$ candidate features per split). Bootstrap-aggregated decision trees that each split on a pixel/feature threshold minimising Gini impurity; prediction is the average class vote.

SVM with RBF kernel (`SVC`). Solves the soft-margin dual with kernel $K(x, x') = \exp(-\gamma\|x - x'\|^2)$, $\gamma = 1/(d\,\text{Var}(X))$ (`gamma='scale'`), one-vs-one for 10 classes. C controls the margin/error trade-off.

Why HOG may help differently. A linear model on pixels can only weigh each pixel independently, so any neighbourhood interaction (an edge) has to be created by the descriptor. A kernel SVM or a forest can construct such interactions itself, so we expect the benefit of HOG to shrink with model flexibility. This hypothesis is tested below.

Metrics. Accuracy; macro-F1 $= \frac{1}{10}\sum_c F1_c$; per-class recall $R_c = TP_c/(TP_c + FN_c)$; parameter count (definition per family, Table II); inference time per image (single-threaded `predict` on the 10,000 test images) plus feature-extraction time.

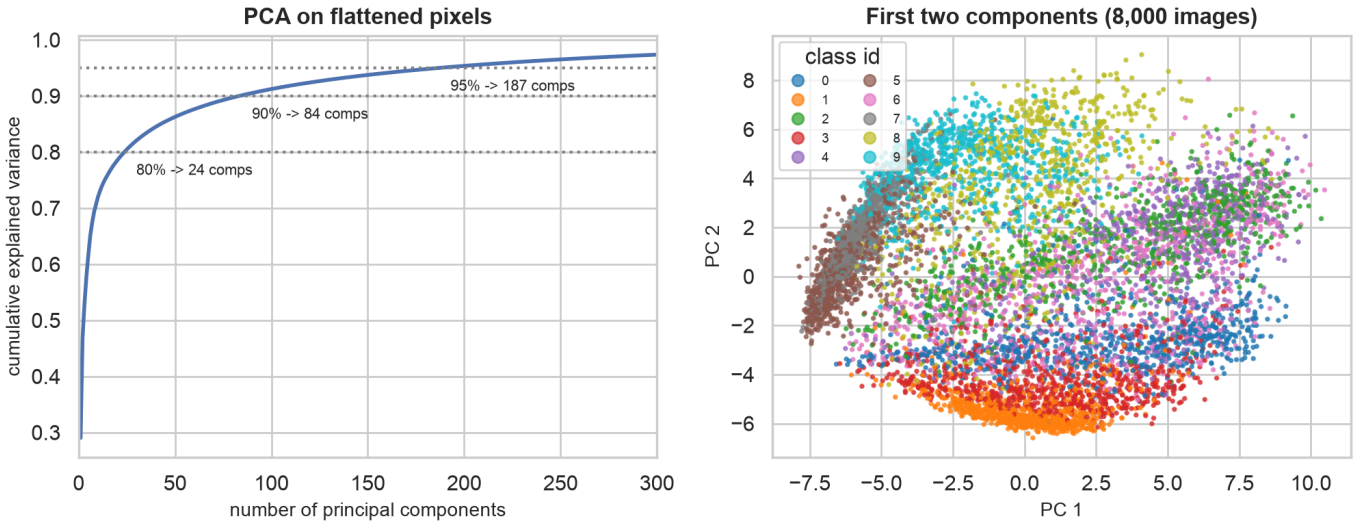


Fig. 8. PCA: cumulative variance and first two components.

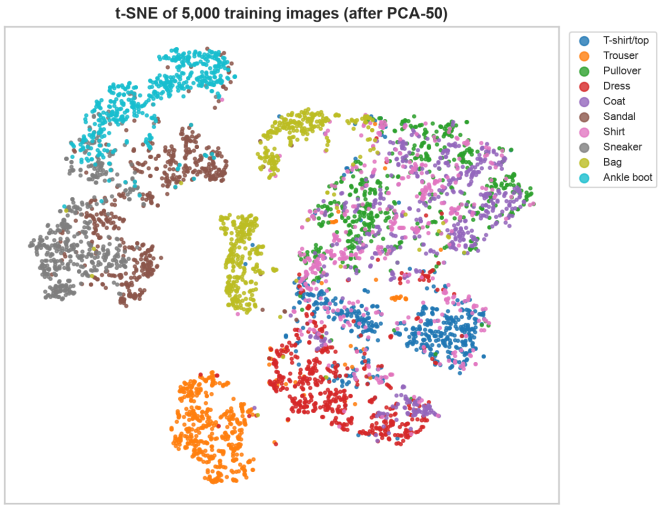


Fig. 9. t-SNE of 5,000 training images.

TABLE II
DEFINITION OF "PARAMETERS".

Model	Counted numbers
LogReg	weights + biases
kNN	stored training matrix
Random Forest	2 per internal node (feature, threshold)
SVM-RBF	support vectors + dual coefficients + intercepts

VI. EXPERIMENTAL PROTOCOL

1) **Split.** The official 60,000/10,000 split is untouched. A stratified 20% of the training set (12,000 images, `train_test_split`, seed 42) forms the validation set; 48,000 images remain for fitting.
2) **Tuning.** Each model's small grid (Table III) is scored on validation accuracy. The SVM grid is fitted on a stratified 15,000-image subsample of the 48,000 because of its cubic worst-case cost, but scored on the full validation set.
3) **Final fit.** The chosen configuration is refit on train+validation (60,000 images) and evaluated once on the test set.
4) **Fairness.** Identical seed, scaling policy and tuning budget for all feature sets; the validation set never contains test images; features such as the scaler are fitted on training data only.

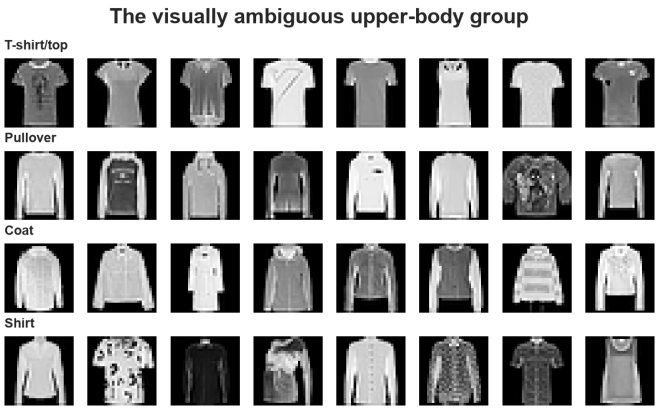


Fig. 10. Upper-body garments: T-shirt/top, Pullover, Coat, Shirt.

TABLE III
HYPER-PARAMETER GRIDS.

Model	Grid
LogReg	$C \in \{0.001, 0.01, 0.1, 1\}$
kNN	$k \in \{3, 5, 9\}$, weights $\in$ {uniform, distance}
RF	depth $\in \{20, \text{None}\}$, min leaf $\in \{1, 2\}$, 300 trees
SVM-RBF	$C \in \{1, 5, 20\}$, $\gamma =$ 'scale'

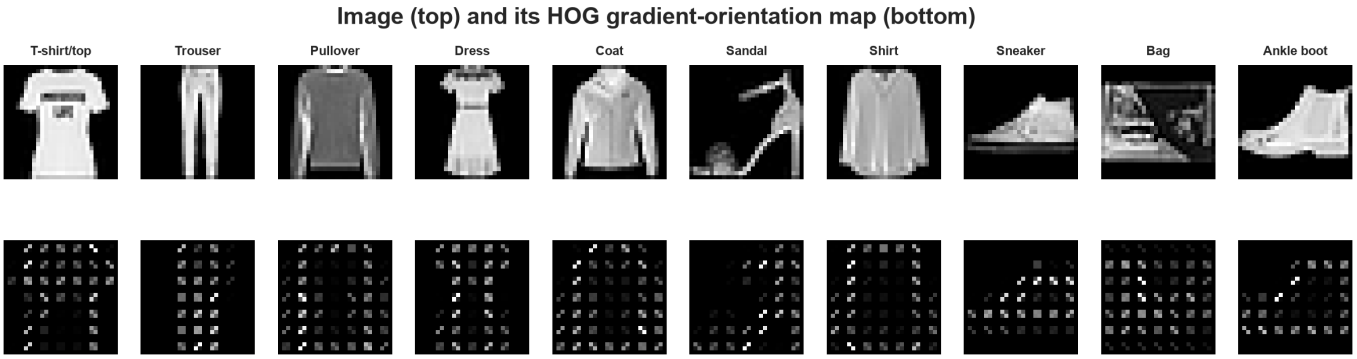


Fig. 11. Images and the HOG orientation map for one example per class.

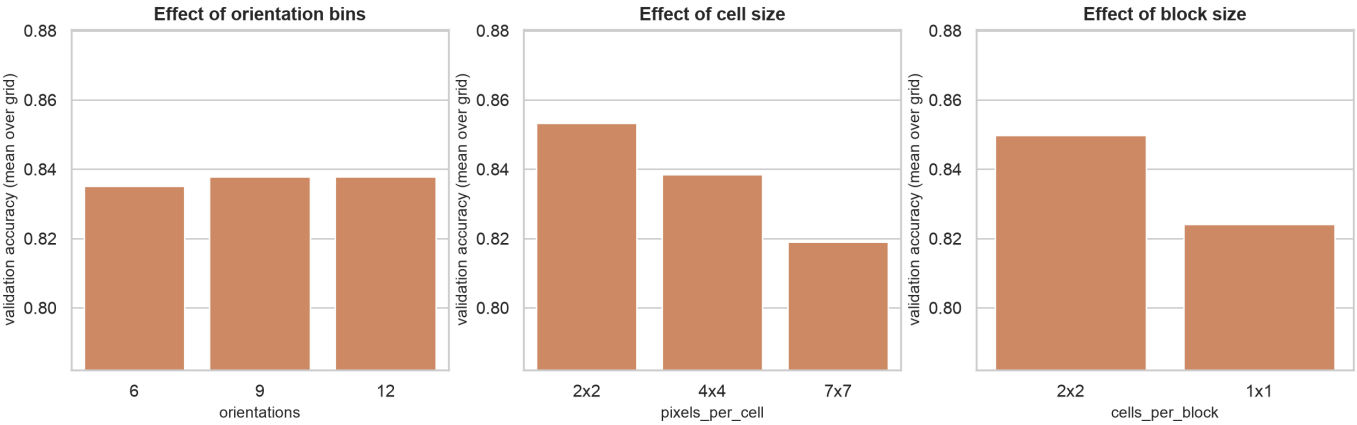


Fig. 12. HOG hyper-parameter ablation (validation accuracy).

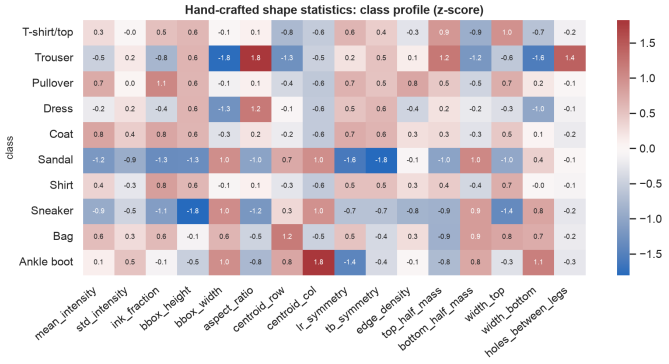


Fig. 13. Class profile of the 16 shape statistics (z-score).

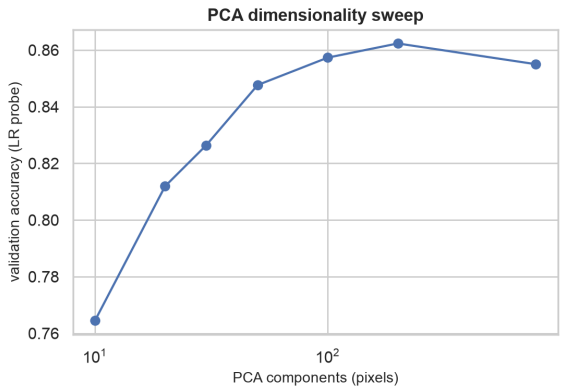


Fig. 14. PCA dimension sweep (LR probe).

VII. Results

Table IV gives the test results; Fig. 15 visualises them.

Does HOG help? Table V reports paired McNemar tests (continuity-corrected χ^2, 1 d.o.f.) between pixels and HOG for each learner. HOG improves Logistic Regression (+3.96 points), k-NN (+2.13) and SVM (+1.42) significantly, while Random Forest is statistically unchanged (+0.21, $p = 0.49$). This supports the hypothesis that the benefit shrinks with model flexibility, and the Random Forest, which can threshold single pixels hierarchically, gains nothing. The 95% bootstrap intervals (1,000 resamples) for the best model are [0.908, 0.919] (HOG+stats) and [0.907, 0.919] (HOG), clearly above the best pixel model [0.893, 0.905].

HOG+stats. Adding the 16 statistics helps Logistic Regression (+0.4) and Random Forest (+0.4) but is neutral for kNN and SVM (+0.07); gains are within bootstrap noise, so we regard HOG alone as the default. Before normalising the

TABLE IV
TEST-SET RESULTS (10,000 IMAGES). LATENCY IS PREDICT TIME PER IMAGE, SINGLE THREAD; HOG EXTRACTION ADDS 0.14 MS (HOG) OR 0.19 MS (HOG+STATS) PER IMAGE.

Model	Features	Val acc	Test acc	Macro-F1	Fit (s)	Infer (ms)	Parameters
LogReg	pixels	0.8633	0.8456	0.8447	8.8	0.0013	7,850
LogReg	HOG	0.8913	0.8852	0.8848	6.1	0.0014	8,650
LogReg	HOG+stats	0.8963	0.8894	0.8890	7.8	0.0013	8,810
kNN	pixels	0.8592	0.8561	0.8564	0.1	0.316	47.0M
kNN	HOG	0.8821	0.8774	0.8777	0.0	0.271	51.8M
kNN	HOG+stats	0.8826	0.8776	0.8780	0.0	0.313	52.8M
RF	pixels	0.8862	0.8760	0.8746	22.8	0.037	2.54M
RF	HOG	0.8823	0.8781	0.8769	39.5	0.042	3.23M
RF	HOG+stats	0.8848	0.8821	0.8811	41.3	0.041	2.51M
SVM-RBF	pixels	0.8888	0.8988	0.8985	192	4.53	15.0M
SVM-RBF	HOG	0.8990	**0.9130**	**0.9130**	207	5.36	18.1M
SVM-RBF	HOG+stats	0.9008	**0.9137**	**0.9137**	223	5.28	17.5M

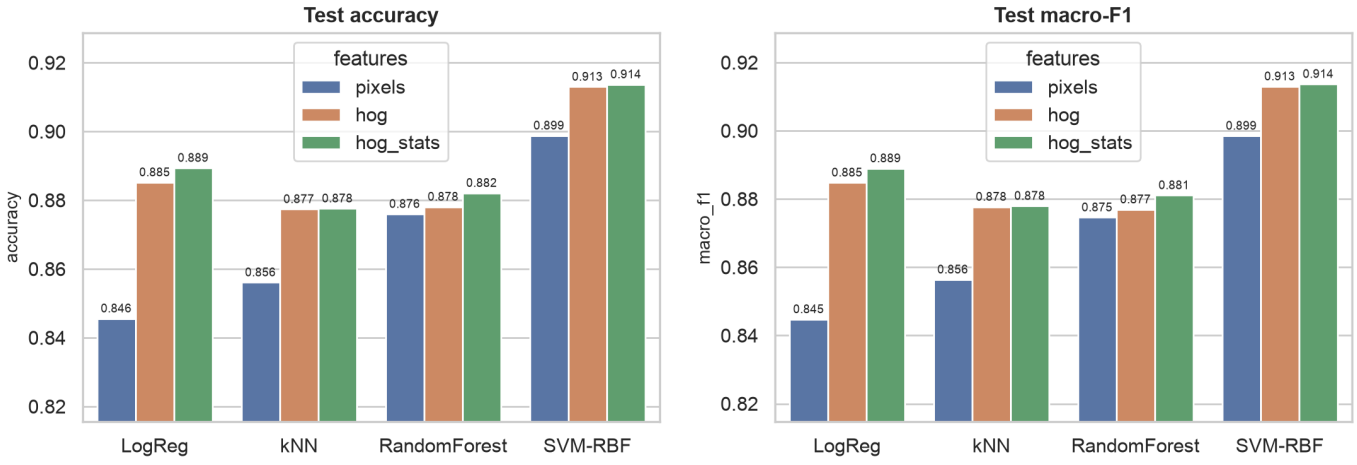


Fig. 15. Test accuracy and macro-F1 for all configurations.

TABLE V
MCNEMAR TEST, PIXELS VS. HOG.

Model	Δacc	only-pix	only-HOG	χ^2	p
LogReg	+0.0396	411	807	128.1	$< 10^{-6}$
kNN	+0.0213	515	728	36.2	2×10^{-9}
RF	+0.0021	404	425	0.48	0.49
SVM	+0.0142	345	487	23.9	10^{-6}

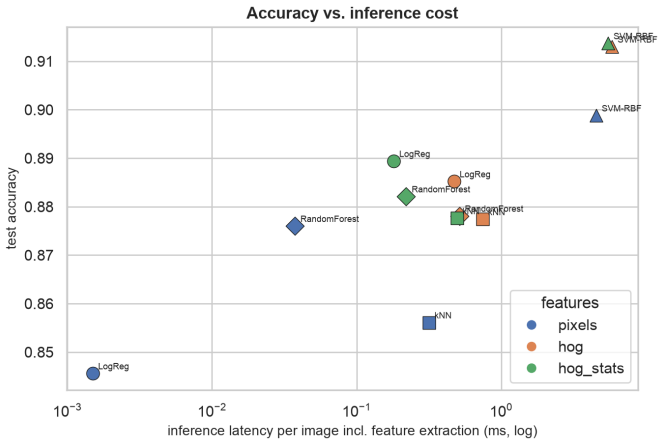


Fig. 16. Accuracy versus inference latency.

statistics to $[0, 1]$, the SVM on HOG+stats scored only 0.8868, showing that unscaled heterogeneous features can dominate a kernel's distance.

Data efficiency. With a fixed Logistic Regression (Fig. 17), HOG is better at every size, and at 5,000 images it already matches pixels at 20,000 (0.863 vs. 0.858).

Cost. Fig. 16 shows accuracy against latency. Logistic Regression is ∼4,000× faster than the SVM at inference and 2,000× smaller, at a cost of 2.8 points. kNN stores 47–53 million numbers. The Random Forest is a good middle option at 0.04 ms but gains nothing from HOG.

Per-class recall. Table VI lists the best pixel and best HOG models. Trouser, Bag, Sandal, Sneaker and Ankle boot exceed 0.96 recall for both, while Shirt (0.705 → 0.750), Pullover (0.830 → 0.856), Coat (0.838 → 0.875) and T-shirt (0.862 →

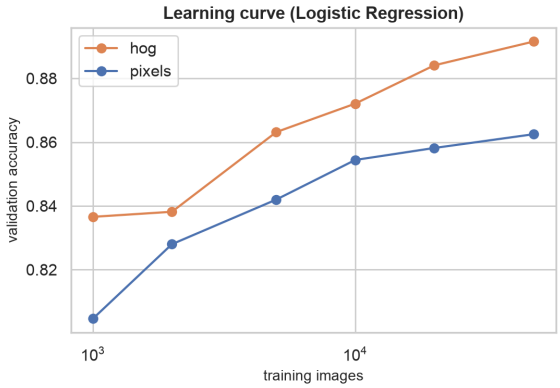


Fig. 17. Learning curves (Logistic Regression).

TABLE VI
PER-CLASS RECALL OF SVM-RBF.

Class	Pixels	HOG
T-shirt/top	0.862	0.873
Trouser	0.970	0.975
Pullover	0.830	0.856
Dress	0.911	0.918
Coat	0.838	0.875
Sandal	0.968	0.971
Shirt	0.705	0.750
Sneaker	0.966	0.967
Bag	0.979	0.981
Ankle boot	0.959	0.964

0.873) are the weak classes; HOG helps these most.

VIII. Error Analysis

Fig. 19 shows the confusion matrices of the best pixel and the best HOG model (both SVM-RBF), and Fig. 20 their difference. The largest confusions for the pixel model are Shirt→T-shirt (120), T-shirt→Shirt (91), Shirt→Pullover (83), Coat→Pullover (79) and Pullover→Coat (77), exactly the pairs predicted by the mean-image distances in Section III.

The share of all errors that fall inside the upper-body group (T-shirt, Pullover, Coat, Shirt) is 58–65% for every model: 64.8% (SVM pixels, 656 of 1,012 errors) and 63.4% (SVM HOG, 552 of 870). HOG removes 142 errors in total, 104 of them in this group, but it cannot remove the ambiguity because at 28×28 the discriminating details of a shirt (collar, buttons, cuffs) are one or two pixels wide. Fig. 21 shows misclassified examples: several are ambiguous even for a human.

IX. Discussion, Limitations and Phase 2

Answer to the research question. A fixed HOG descriptor is a cheap (0.14 ms) and statistically significant improvement for linear, instance-based and kernel learners, and the best classical model is a kernel SVM on HOG (91.3%). It does not help a Random Forest. Flattened pixels are not always worse; they are worse when the learner cannot build neighbourhood interactions itself.

Limitations. (i) The HOG configuration was selected with a Logistic-Regression probe, not per classifier. (ii) A single validation split is used; test uncertainty is bootstrap-based but there is no repeated cross-validation. (iii) The SVM is tuned on a 15,000-image subsample. (iv) Grids are small. (v) The 28×28 grayscale resolution discards the texture that separates shirts from T-shirts.

Phase 2. A two- or three-block CNN with pooling, dropout and early stopping will be trained on the *same* splits and compared on the same metrics. The CNN can learn its own hierarchical filters and may resolve the upper-body group better than any fixed descriptor.

X. Conclusion

We built a reproducible Phase-1 pipeline for Fashion-MNIST: audit and EDA, three feature sets, four classifiers, strict validation, paired significance tests and error analysis. HOG improves three of four classifiers significantly; the best model reaches 91.3% accuracy and 0.913 macro-F1; and the Shirt/T-shirt/Pullover/Coat group dominates the remaining errors.

Reproducibility and Code

All code, notebooks, the Streamlit presentation and raw result tables are in the project repository (`README.md`). Seeds are fixed (42); `scripts/run_all.sh` regenerates every figure and table.

Appendix A
Software Components

TABLE VII
MAIN FUNCTIONS USED.

Function / class	Purpose
`gzip`, `np.frombuffer`	read IDX files
`train_test_split` (stratify)	validation split from train
`skimage.feature.hog`	HOG descriptor
`scipy.ndimage.sobel`	edge density
`PCA`, `TSNE`	dimensionality reduction/visualisation
`f_classif`	pixel F-score
`StandardScaler`, `Pipeline`	scaling inside the model
`LogisticRegression`, `KNeighborsClassifier`, `RandomForestClassifier`, `SVC`	the four learners
`accuracy_score`, `f1_score`, `recall_score`, `confusion_matrix`	metrics
`scipy.stats.ks_2samp`, `chi2`	KS test, McNemar test
`joblib`	parallel feature extraction, model saving
`matplotlib`, `seaborn`, `plotly`, `streamlit`	figures and presentation app

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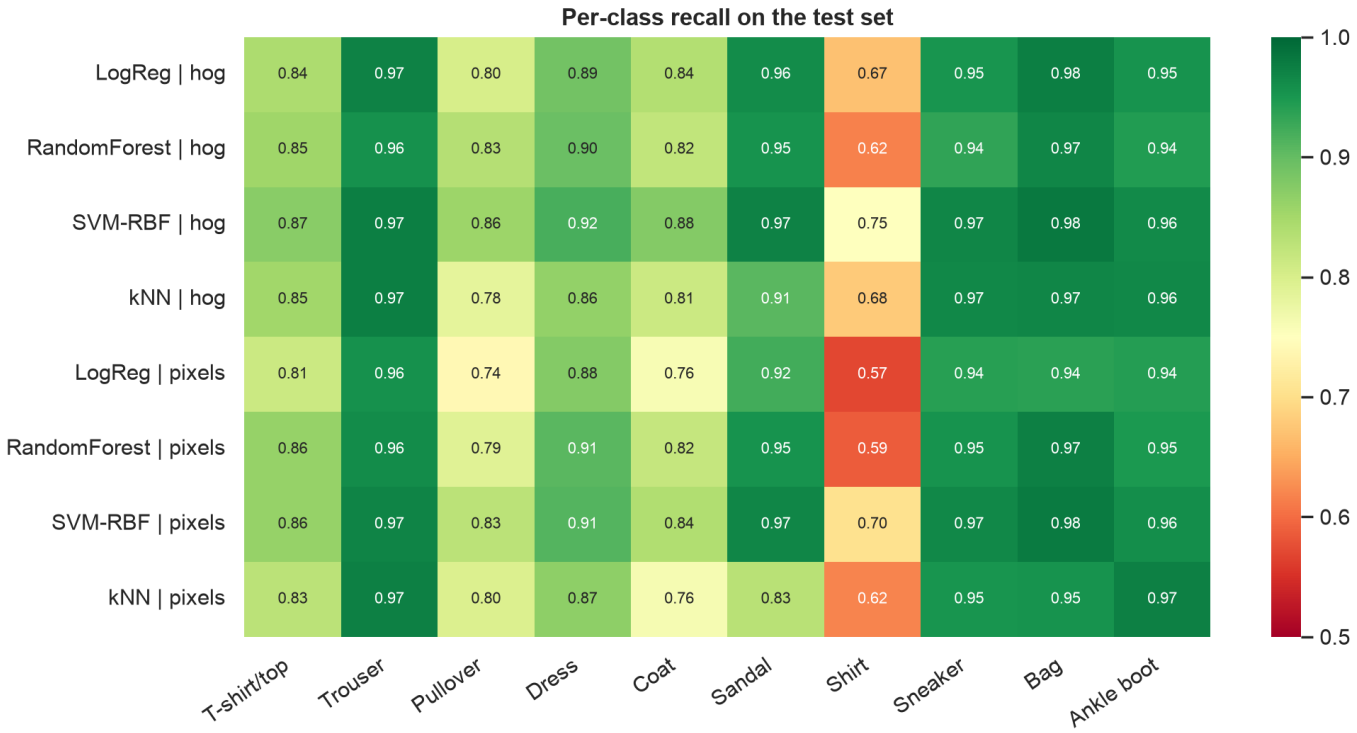


Fig. 18. Per-class test recall for all models (pixels and HOG).

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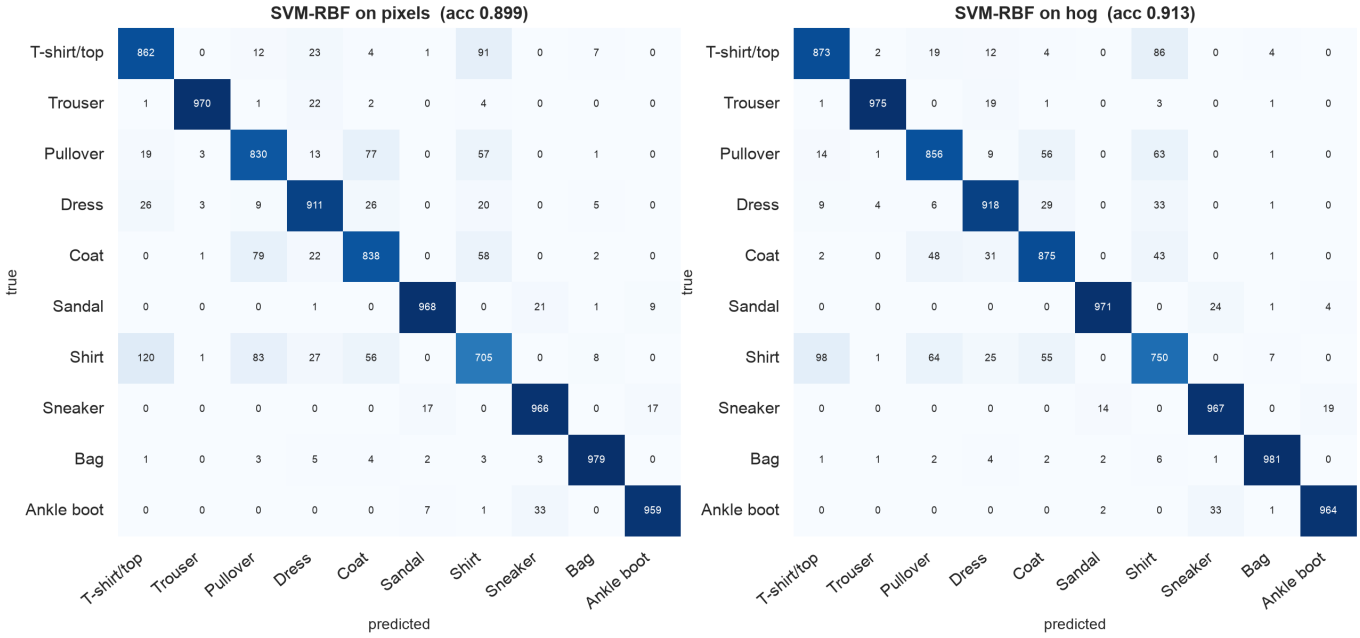


Fig. 19. Confusion matrices: best pixel model vs. best HOG model.

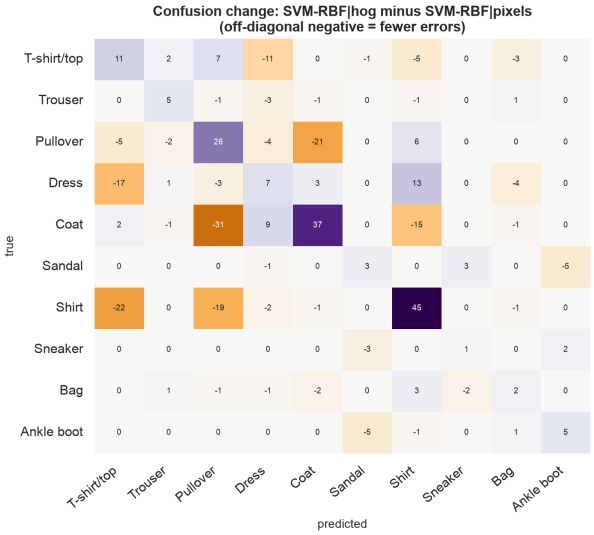


Fig. 20. Confusion change (HOG minus pixels).

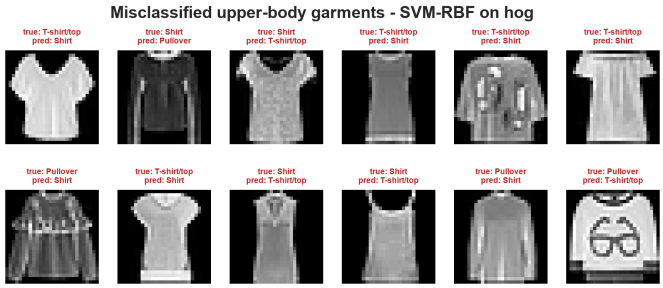


Fig. 21. Misclassified upper-body garments (SVM, HOG).